

SpacECE Smart Parenting Portal

1. System Overview & Technology Stack

The **SpacECE Smart Parenting & Early Childhood Education Portal** is an end-to-end full-stack MERN application (MongoDB, Express.js, React.js, Node.js) designed for interactive early childhood learning, parent diagnostic monitoring, and educator analytics aligned with **NEP 2020 benchmarks** across 5 primary developmental domains (*Literacy, Numeracy, Cognitive, Creativity, Emotional*).

2. Visual Codebase Directory & Module Hierarchy Tree

SpacECE MongoDB — Complete Visual File Skeleton & Module Tree

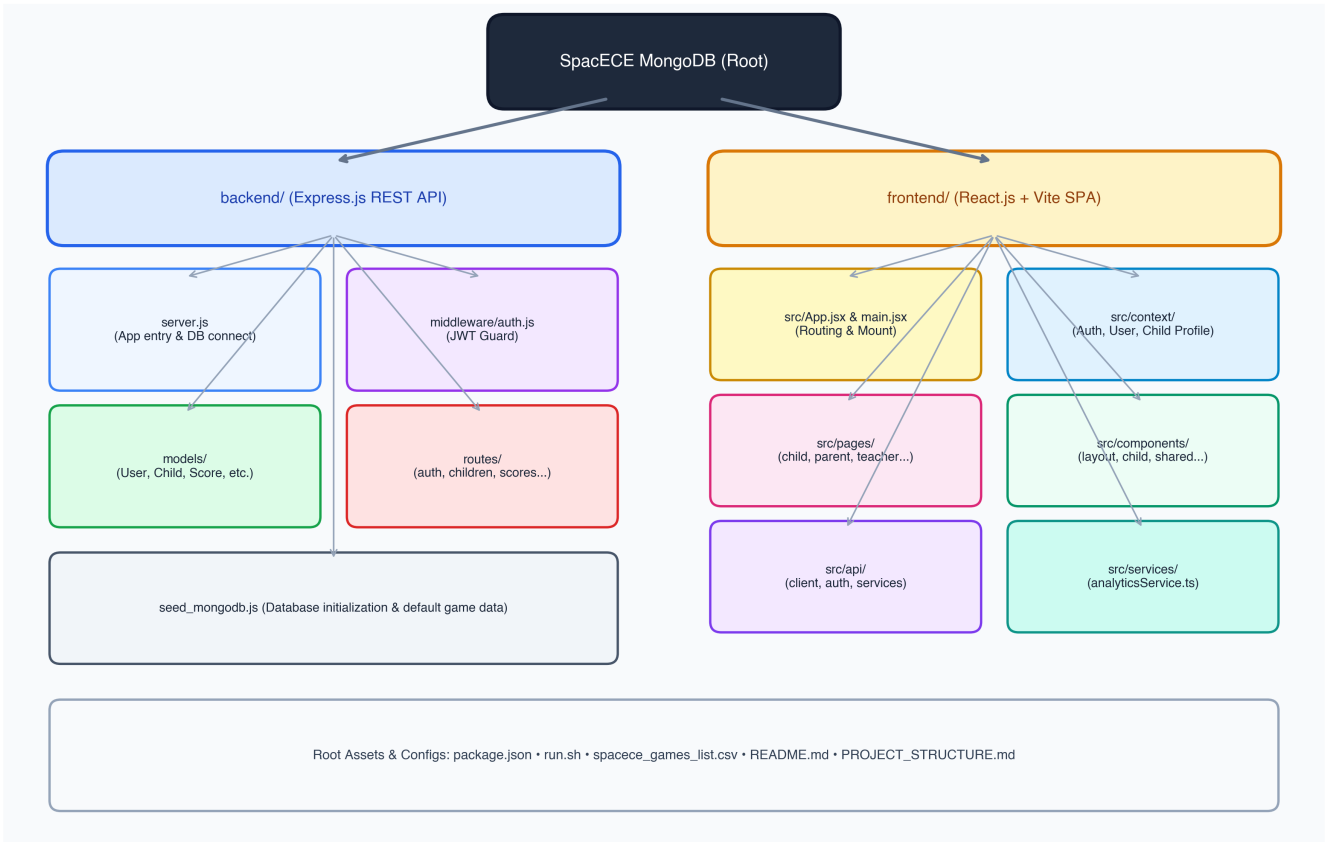
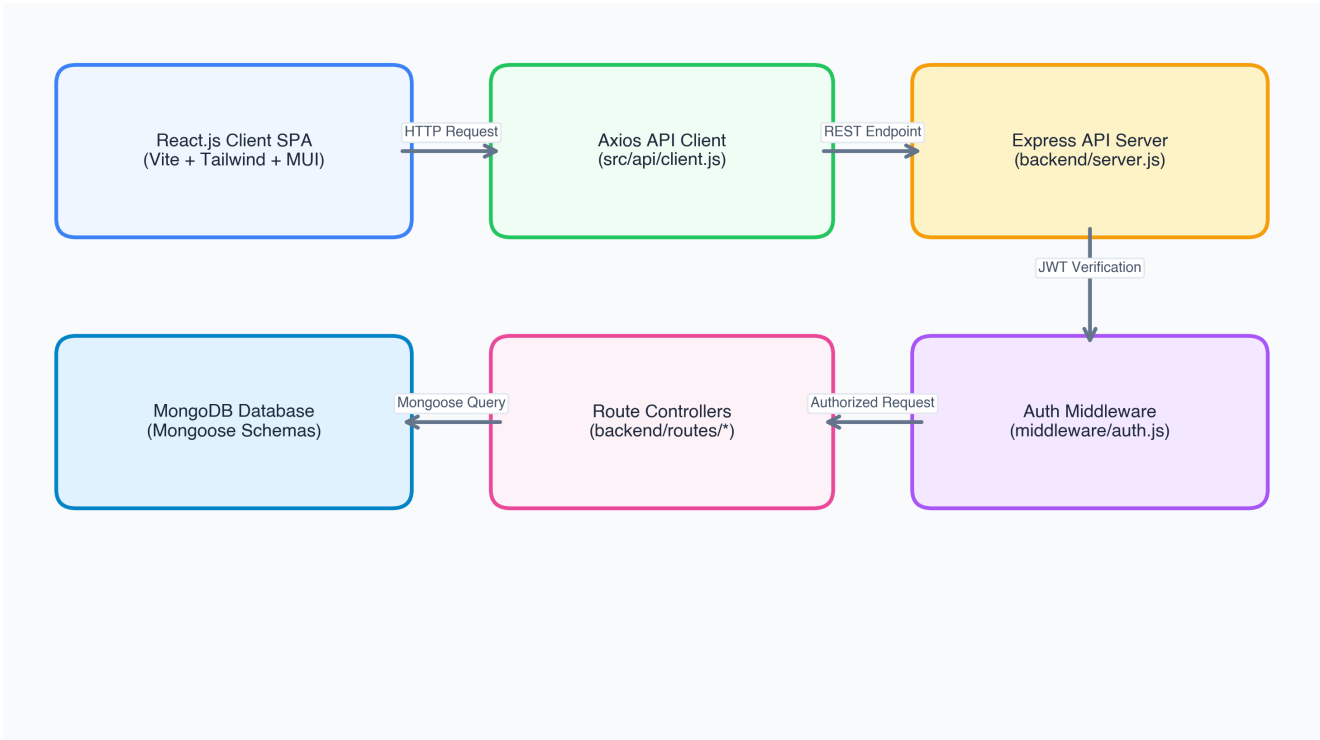


Diagram Interpretation: The project root contains dual sub-applications: `/backend` (Express REST API with Mongoose ORM) and `/frontend` (Vite-bundled React SPA). Root configuration scripts like `run.sh` manage parallel server execution.

3. High-Level Multi-Tier Architecture & Request Pipeline

SpacECE Smart Parenting Portal — System Architecture & Data Flow



Component Interaction Flow: Browser requests are captured by React Router views, passed through React Contexts (AuthContext, UserContext), dispatched via Axios client with JWT headers, verified by Express Auth Middleware, handled by Route Controllers, and executed against MongoDB collections.

4. Backend Subsystem Matrix (`/backend`)

Database Collections & Schemas (`backend/models`)

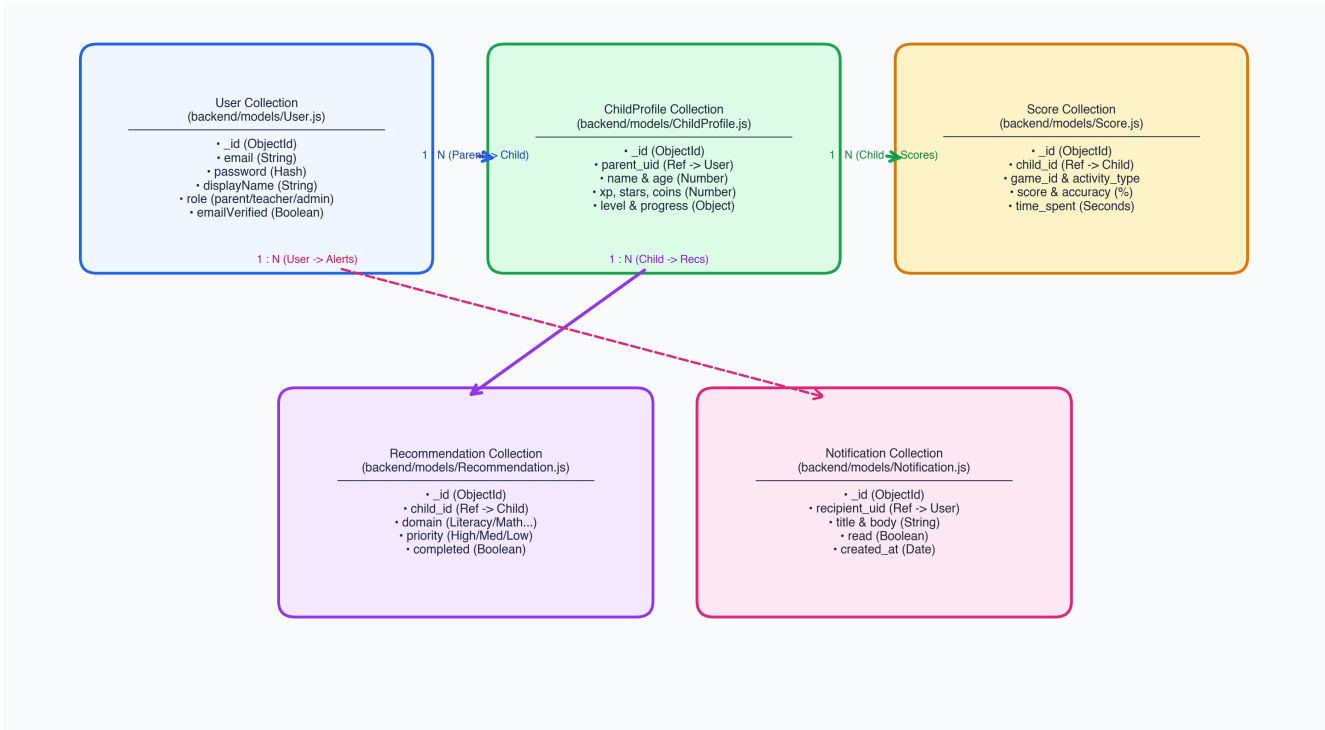
Model File	Collection	Schema Responsibilities & Attributes
User.js	users	User accounts: email, bcrypt password hash, role (parent/teacher/admin), emailVerified.
ChildProfile.js	childprofiles	Child profile state: name, age, parent_uid link, xp, stars, coins, level, mascot, module progress.
Score.js	scores	Performance logs: child_id link, game_id, activity_type, score, accuracy (%), time_spent.
Literacy.js	literacies	Literacy game configurations: phonics, stories, word builder, sight words.
CognitiveSel.js	cognitivesels	Cognitive & SEL game configurations: brain world, emotion check-in, branching choice stories.
Recommendation.js	recommendations	AI & Educator recommendations: child_id, domain, reason, priority (High/Med/Low), status.
Notification.js	notifications	Alert feed items: recipient_uid, title, body, read status, creation timestamp.
Center.js	centers	Early Childhood Learning Center directory entries.

Express API Routers (backend/routes/)

Router File	Base Path	API Endpoints & Handlers
auth.js	/api/auth	POST /register, /login, /google, GET /me, PUT /change-password, /change-email.
children.js	/api/children	GET/POST /children, GET/PUT /children/:id (profile CRUD & awardProgress).
scores.js	/api/scores	POST /scores (submit score), GET /scores (fetch domain scores for analytics).
literacy.js	/api/literacy	GET /literacy/games, GET /literacy/stories (phonics, story reader content).
cognitiveSel.js	/api/cognitive-sel	GET /cognitive-sel/games, /branching-stories (SEL & emotion content).
milestones.js	/api/milestones	GET/POST /milestones/recommendations (NEP 2020 milestone activity focus).
notifications.js	/api/notifications	GET /notifications, PUT /notifications/:id/read (alert feed updates).

5. MongoDB Database Entity-Relationship (ERD) Diagram

MongoDB Database Entity-Relationship & Schema Diagram (ERD)



Collection Relationships: User maintains a 1:N relationship with ChildProfile records via `parent_uid`. Each ChildProfile links to multiple Score logs and Recommendation action items. System notifications map directly to target user IDs.

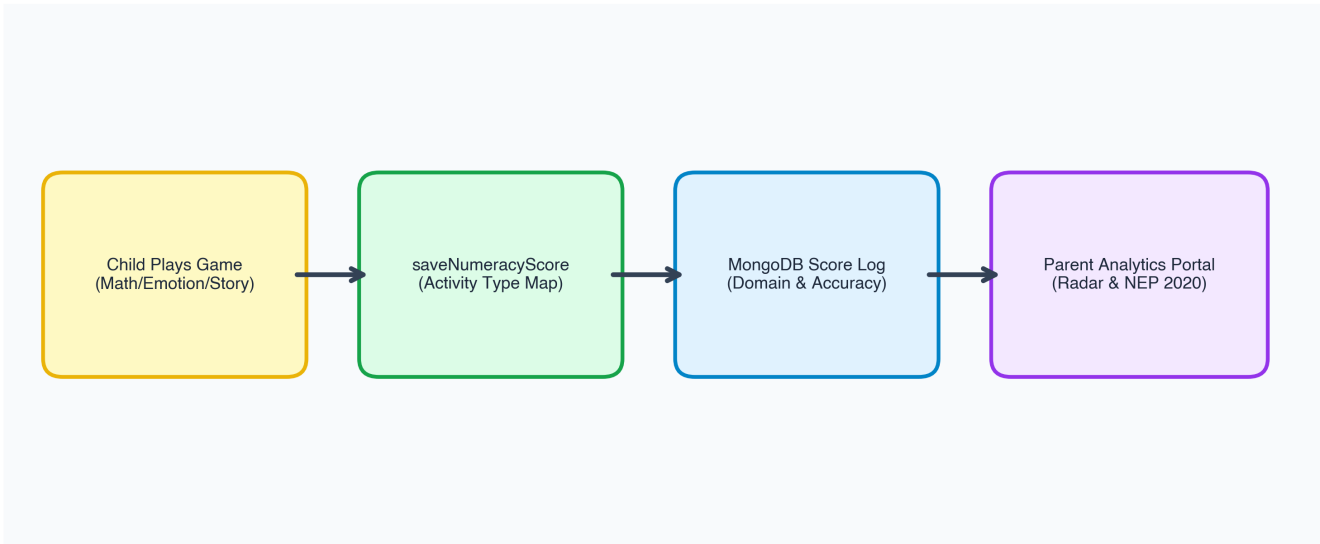
6. Frontend Subsystem & Page Matrix (`/frontend/src`)

React View Pages Breakdown (`src/pages/`)

Domain Directory	Key Page Components	Responsibility & Learning Objectives
auth/	Login, Register, RoleSelection, Terms, Privacy	User sign-in, account creation with role pre-selection (Parent/Teacher), legal pages.
child/	Home (Dashboard), Adventure, AvatarPage, Awards	Child portal dashboard, 3D island map explorer, avatar customization, reward trophies.
numeracy/	MathWorld, PuzzleWorld, LogicIsland, NumberAdventure	Counting, shape side identification, size comparison, number matching, arithmetic.
cognitive-sel/	EmotionWorldPage, StoryWorldPage, SpatialBrainPage	Emotion check-in, face recognition, friendship social stories, branching narratives.
literacy/	ReadingWorldPage, PhonicsLandPage, StoryWorldPage	Letter phonics sounds, sight words, picture matching, independent story reader.
analytics/	ParentDashboard, DevelopmentalLearningCurves	NEP 2020 domain competency radar chart, time allocation bars, school readiness score.
parent/ & teacher/	ParentDashboard, TeacherDashboard, ChildrenList	Child profile management, classroom roster monitoring, assigning activity recommendations.
settings/	AccountSettings, ProfileSettings	Parent & educator account credentials, display name updates, email verification.

7. Game Score Logging & Analytics Pipeline

End-to-End Game Score Logging & Analytics Pipeline



Analytics Processing Flow: Activity completion in child games calls `saveNumeracyScore` which categorizes games into 5 explicit domains (*Literacy, Numeracy, Cognitive, Creativity, Emotional*). Scores logged to MongoDB are fetched by `analyticsService.ts` to calculate readiness scores and plot radar charts.

8. Platform Role & Permission Access Matrix

Role	Primary Access Paths	Capabilities & System Permissions
Child	/child/*, /math-world, /child/emotion-world	Play educational games, gain XP/stars/coins, unlock island paths, customize avatar.
Parent	/parent/*, /analytics/*, /settings/*	Manage child profiles, view NEP 2020 milestone analytics, complete action items.
Teacher	/teacher/*, /analytics/teacher	Track classroom rosters, diagnose learning delays, assign activity recommendations.
Admin	/admin/*	Platform user management, system metrics oversight, configure alert thresholds.